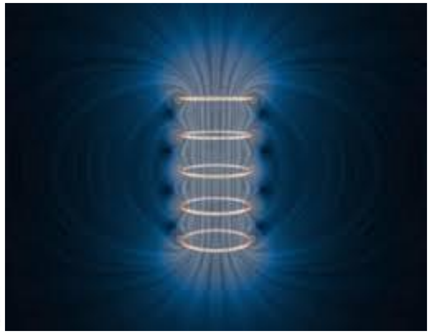


# MAGNETIC CIRCUITS

# Objectives

- To conceptualize the basic terms associated with magnetic circuits.
- To analogize electrical and magnetic circuits.
- To study the Composite Magnetic Circuit.

# Contents

- Introduction
- Basic definitions
- Magnetic circuit
- Composite Magnetic Circuit.

# Introduction

- An electrical machine is a device which converts electrical power (voltages and currents) into mechanical power(torque and rotational speed), and/or vice versa.
- A motor describes a machine which converts electrical power to mechanical power; a generator (or alternator) converts mechanical power to electrical power.
- Almost all practical motors and generators convert energy from one form to another through the action of a magnetic field.
- Transformers are usually studied together with generators and motors because they operate on the same principle, the difference is just in the action of a magnetic field to accomplish the change in voltage level.2

# Principle of Electromagnet

- The principles of magnetism play an important role in the operation of an electric machine.
- The basic idea behind an electromagnet is a magnetic field around the conductor can be produced when current flows through a conductor. In other word, the magnetic field only exists when electric current is flowing
- By using this simple principle, you can create all sorts of things, including motors, solenoids, read/write heads for hard disks and tape drives, speakers, and so on.

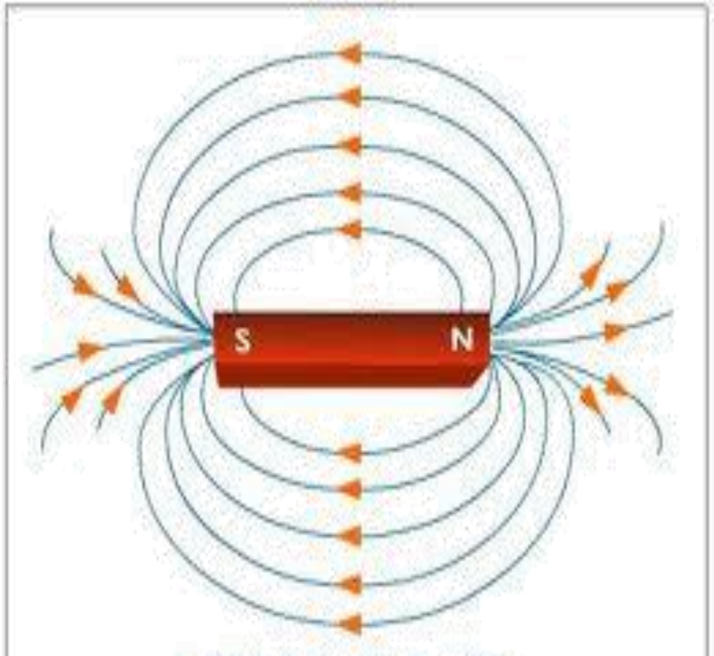
# Introduction

## Magnetic lines of force:

Closed path radiating from north pole, passes through the surrounding, terminates at south pole and is from south to north pole within the body of the magnet.

### Properties:

- Each line forms a closed loop and never intersect each other.
- Lines are like stretched elastic cords.
- Lines of force which are parallel and in the same direction repel each other.




Field Lines Around a Bar Magnet

# Introduction

## Magnetic Field

- The space around which magnetic lines of force act.

- Strong near the magnet and weakens at points away from the magnet.

# Introduction

## Magnetic Materials

**Properties:**

Points in the direction of geometric north and south pole when suspended freely and attracts iron fillings.

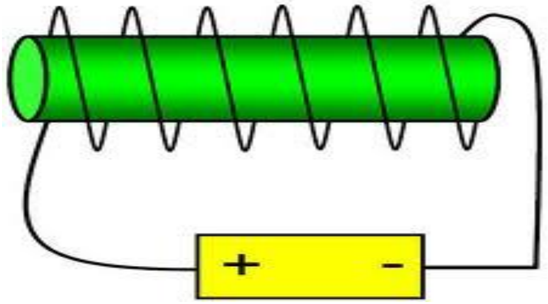
**Classification :**

- Natural Magnets
- Temporary magnets (exhibits these properties when subjected to external force)
- Non-magnetic materials.

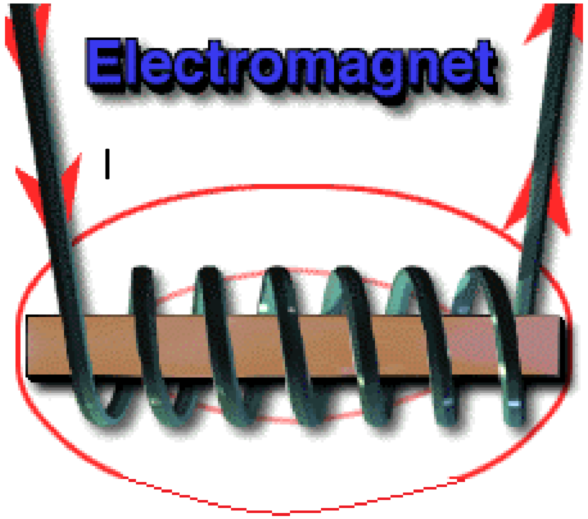
# Introduction

## Electromagnets:

Principle: An electric current flowing in a conductor creates a magnetic field around it.

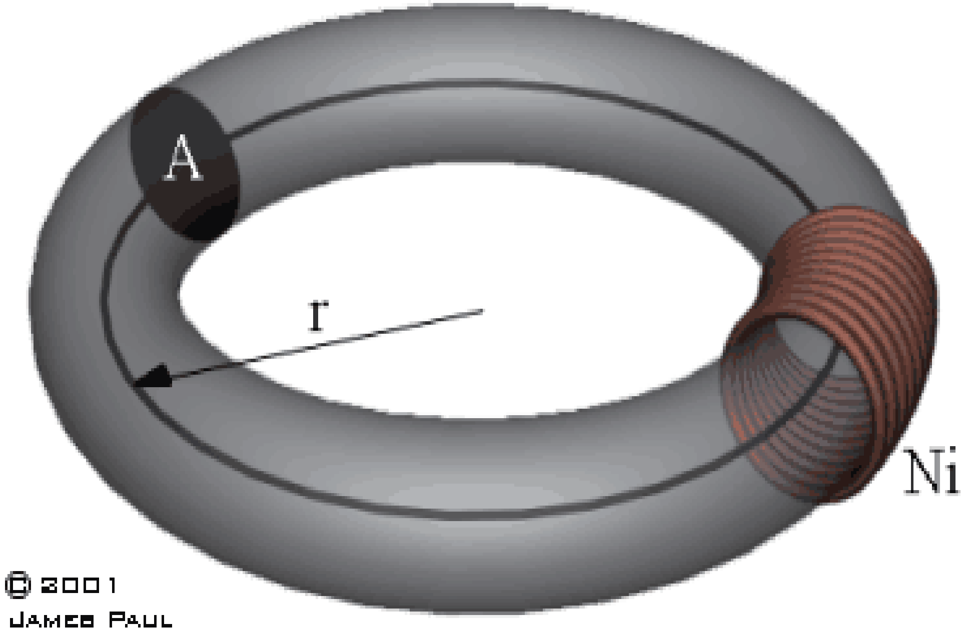


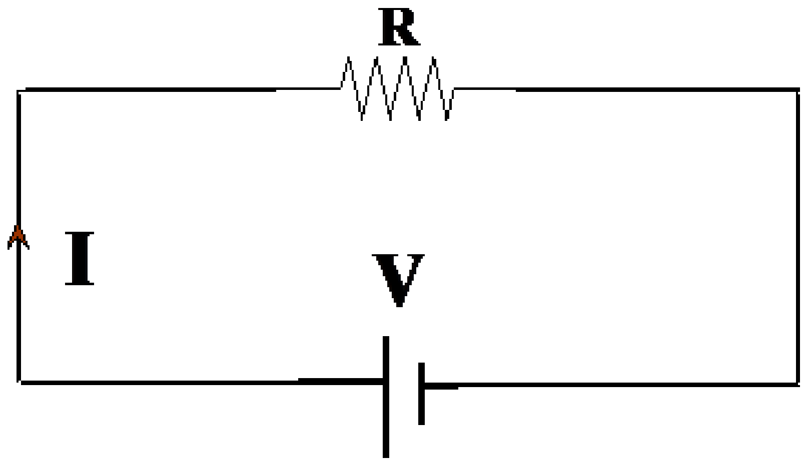

- Strength of the field is proportional to the amount of current in the coil.
- The field disappears when the current is turned off.
- A simple electromagnet consists of a coil of insulated wire wrapped around an iron core.
- Widely used as components of motors, generators, relays etc.



# Magnetic circuit

The complete closed path followed by any group of magnetic lines of flux




Equivalent electrical circuit

# Basic Definitions

## Magneto Motive Force, MMF (*F*)

- Force which drives the magnetic lines of force through a magnetic circuit
- *MMF, F = ΦS*, where 'Φ' is the magnetic flux and 'S' is the Reluctance of the magnetic path.

Analogy: EMF, $V=IR$

- Also, For Electromagnets:

MMF= $N\,I$ (No. of turns*Current),

where N is the number of turns of the coil and I is the current flowing in the coil

- Unit: *AT* (Ampere Turns)

# Basic Definitions

## **Magnetic flux ($\Phi$):**

- Number of magnetic lines of force created in a magnetic circuit.
- Unit : *Weber (Wb)*
- 1 weber = $10^8$ lines of force

Analogy: Electric Current, I

# Basic Definitions

## Reluctance [S]

- Opposition of a magnetic circuit to the setting up of magnetic flux in it.

$$Flux = \phi = BA;\;\; F = mmf = Hl;\;\; B = \mu H$$

$$\frac{\phi}{F} = \frac{BA}{Hl} = \frac{\mu_0 \mu_r A}{l};\;\; Hence \quad \phi = \left(\frac{\mu_0 \mu_r A}{l}\right) F$$

$$\phi = \frac{F}{\left(\dfrac{l}{\mu_0 \mu_r A}\right)} = \frac{F}{S};\;\; \text{where}\;\; \mathrm{S} = \left(\frac{l}{\mu_0 \mu_r A}\right)$$

- $S = F/\phi$

Analogy: Resistance

- Unit: $AT / Wb$

# Basic Definitions

## Magnetic Flux Density (*B*):

- No. of magnetic lines of force created in a magnetic circuit per unit area normal to the direction of flux lines
- $B = \Phi/A$

Analogy: Current Density

- Unit : *Weber/m$^2$ (Tesla)*

## Magnetic Field Strength (*H*)

- The magneto motive force per meter length of the magnetic circuit
- $H = (N\,I) / l$
- Unit : *AT / meter*

Analogy: Electric field strength

# Basic Definitions

## Permeability (µ)

- A property of a magnetic material which indicates the ability of magnetic circuit to carry magnetic flux.
- $\mu = B / H$ Analogy: Conductivity
- Unit: ***Henry / meter***
- Permeability of free space or air or non magnetic material $\mu_0 = 4 * \Pi * 10^{-7}$ *Henry/m*
- Relative permeability, $\mu_r : \mu/\mu_0$

# Magnetic circuit

## Analogy with Electric circuits

| **Similarities:** | | | |
|---|---|---|---|
| **Electric circuit** | | **Magnetic circuit** | |
| **Quantity** | **Unit** | **Quantity** | **Unit** |
| **EMF (E=IR)** | **Volt (V)** | **MMF (F=ϕS)** | **Ampere-turns** |
| **Current (I)** | **Ampere (A)** | **Flux (ϕ)** | **Weber (Wb)** |
| **Current density (J)** | **A/ $m^2$** | **Flux density (B)** | **Wb / $m^2$** or **Tesla** |
| **Resistance (R)** | **Ohm (Ω)** | **Reluctance (S)** | **Ampere-turns/Wb** |
| **Electric field strength (E)** | **Volts/m** | **Magnetic field strength (H)** | **Ampere-turns/m** |
| **Conductivity (σ) σ=l/RA** | **Siemen/m** | **Permeability, μ μ=l/SA** | **Henry/m** |

**’l’ is the length and ‘A ‘is the area of cross section of the conductor**

| Sr. No. | Magnetic Circuit | Electric Circuit |
|---|---|---|
| 1 | The closed path traced by magnetic flux is called as Magnetic circuit | The closed path traced by electric current is called as Electric circuit |
| 2 | Ohm’s law – flux is the ratio of mmf to reluctance | Ohm’s law – current is the ratio of emf to resistance |
| 3 | Flux is lines of forces & passes from N-pole to S-pole in magnetic circuit in Wb | Current is flow of electrons from positive to negative in electric circuit in Amp |
| 4 | MMF is driving force in AmpTurns | EMF is driving force in Volts |
| 5 | Reluctance opposes flow of flux in AT / Wb | Resistance opposes flow of current in ohms |
| 6 | Permeance is reciprocal of reluctance in Wb / AT | Conductance is reciprocal of resistance in mho |
| 7 | Flux density is the ratio of flux per cross sectional area in Wb / $m^2$ | Current density is the ratio of current per cross sectional area in Amp / $m^2$ |
| 8 | Reluctance is directly proportional to length of magnetic circuit | Resistance is directly proportional to length of conductor |
| 9 | Reluctivity = $\frac{1}{\mu}$ in Meter / Henry | Resistivity = ρ in ohm meter |
| 10 | Permeability = μ Henry / meter | Conductivity = σ in mho meter |
| 11 | Permanent magnet gives constant mmf | DC battery gives constant emf |
| 12 | Magnetic field strength is the ratio of mmf per length of magnetic circuit | Electric field strength is the ratio of emf per length of conductor |

# Magnetic circuit

## Differences between electric and magnetic circuits

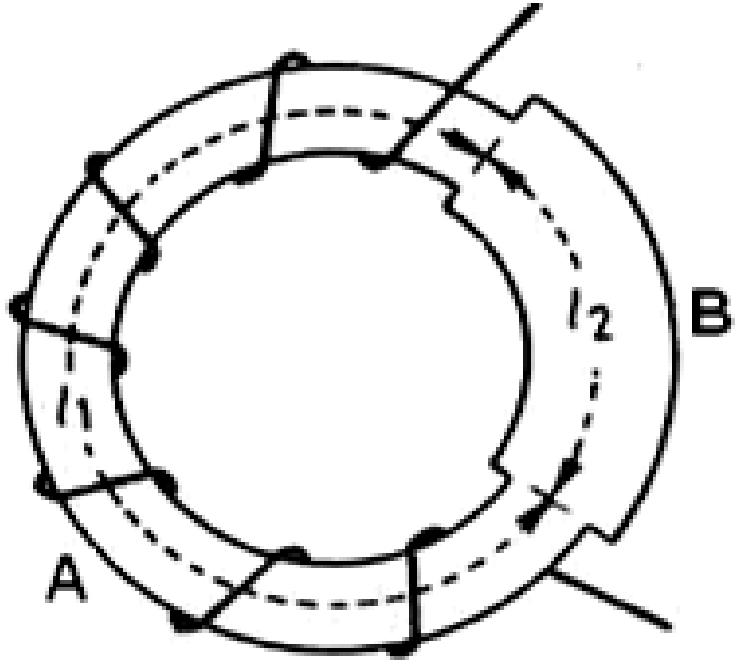
- In electrical circuit current actually flows.
- In magnetic circuit flux is created, and it is not a flow.

# DISSIMILARITY

| Sr. No. | Magnetic Circuit | Electric Circuit |
|---|---|---|
| **1** | Magnetic flux does not flow in circuit | Electric current actually flows in electric circuit |
| **2** | There is no magnetic insulator | There are number of electric insulators |
| **3** | Magnetic lines of forces are closed lines | Electric lines of forces are not closed lines |
| **4** | Magnetic circuit is always a closed circuit | Electric circuit can be closed or open circuit |
| **5** | Magnetic energy is initially needed to create the magnetic flux but not required to maintain it | Electric energy is continuously required to maintain the current |
| **6** | Flux can pass through air also | Current can not pass through air |
| **7** | There is a special quantity $\mu o = 4 \times \pi \times 10^{-7}$ H / m | There is no quantity analogous to $\mu o$ |

# Composite Magnetic Circuit

Case 1 :




$$\mathcal{R}_1 = \frac{l_1}{\mu_1 A_1}$$

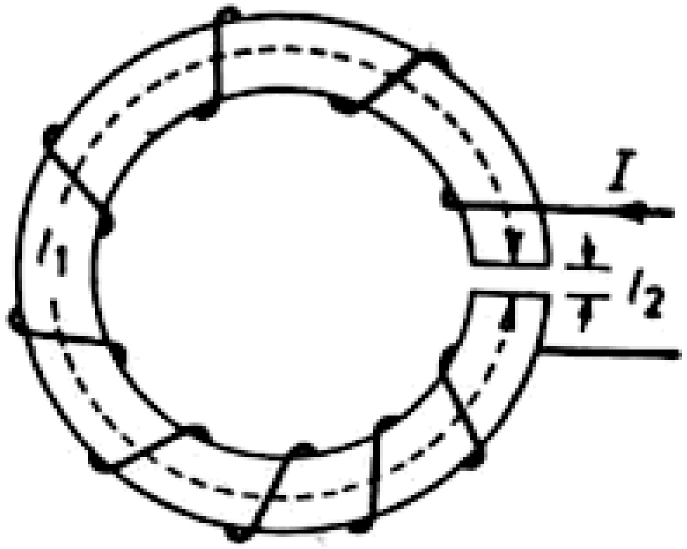
$$\mathcal{R}_2 = \frac{l_2}{\mu_2 A_2}$$

$$\therefore \text{ Total Reluctance, } \mathcal{R} = \mathcal{R}_1 + \mathcal{R}_2 = \frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2}$$

$$\therefore \ \text{Total flux}, \Phi = \frac{\text{mmf of coil}}{\text{total reluctance}}$$

$$= \frac{\mathcal{F}}{\mathcal{R}} = \frac{NI}{\dfrac{l_1}{\mu_1 A_1} + \dfrac{l_2}{\mu_2 A_2}}$$

## Case 2 : (with air gap)




Total reluctance,

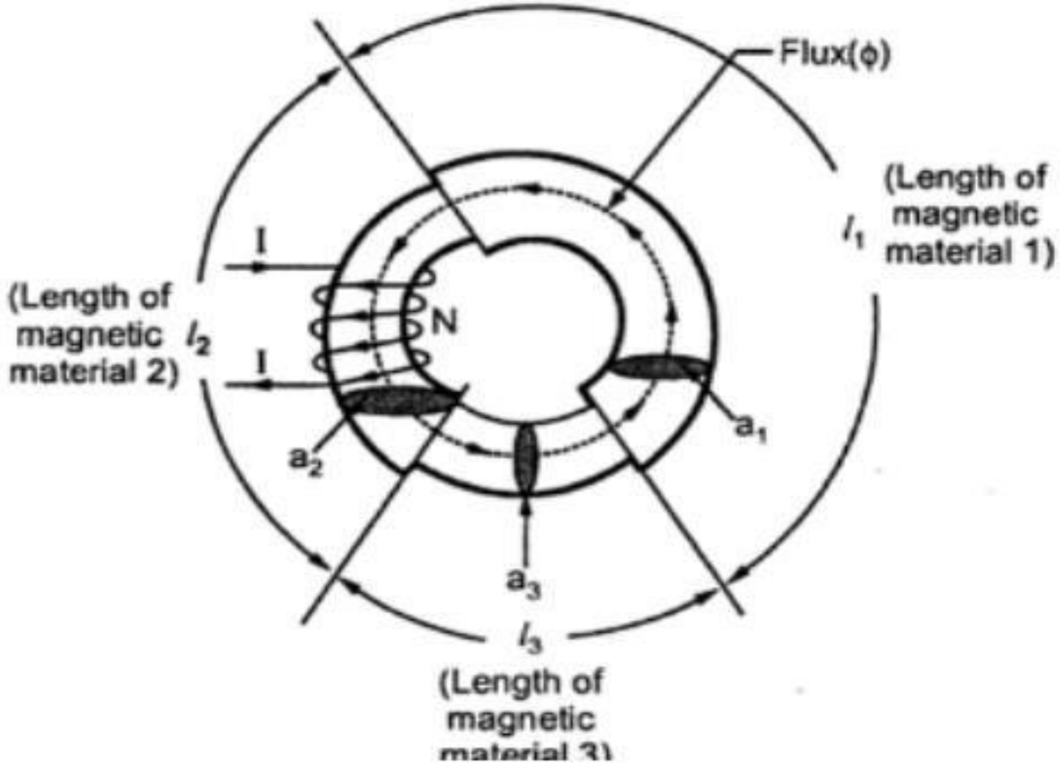
$$\mathcal{R} = \frac{l_1}{\mu_1 A} + \frac{l_2}{\mu_0 A}$$

$$= \frac{1}{\mu_0 A}\left(\frac{l_1}{(\mu_1/\mu_0)} + l_2\right)$$

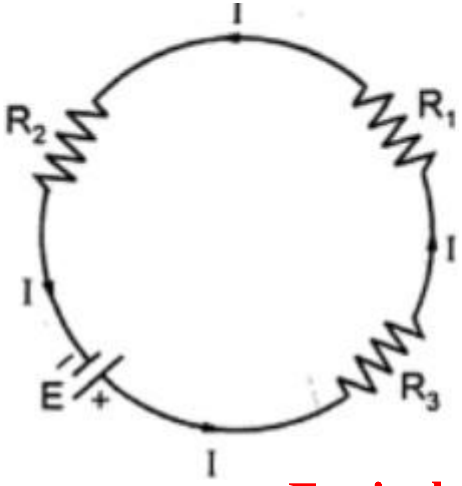
$$= \frac{1}{\mu_0 A}\left(\frac{l_1}{\mu_r} + l_2\right)$$

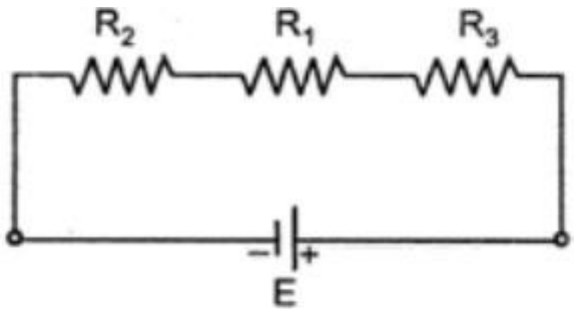
# Series magnetic circuits

- Magnetic circuit composed of various materials of different permeabilities.




- When composite magnetic circuit parts are connected one after the other the circuit is called series magnetic circuit.
- Consider a circular ring made up of different materials of lengths $l_1$, $l_2$ $and$ $l_3$and with cross sectional areas $a_1$, $a_2$ $and$ $a_3$ with absolute permeabilities $\mu_1, \mu_2$ and $\mu_3$.




**Equivalent electric circuit**

$$NI = S_T \phi = (S_1 + S_2 + S_3)\,\phi$$

$$NI = S_1\phi + S_2\phi + S_3\phi$$

$$(m.m.f.)T = (m.m.f.)_1 + (m.m.f.)_2 + (m.m.f.)_3$$

The total m.m.f. also can be expressed as.

$$(m.m.f.)T = H_1 l_1 + H_2 l_2 + H_3 l_3$$

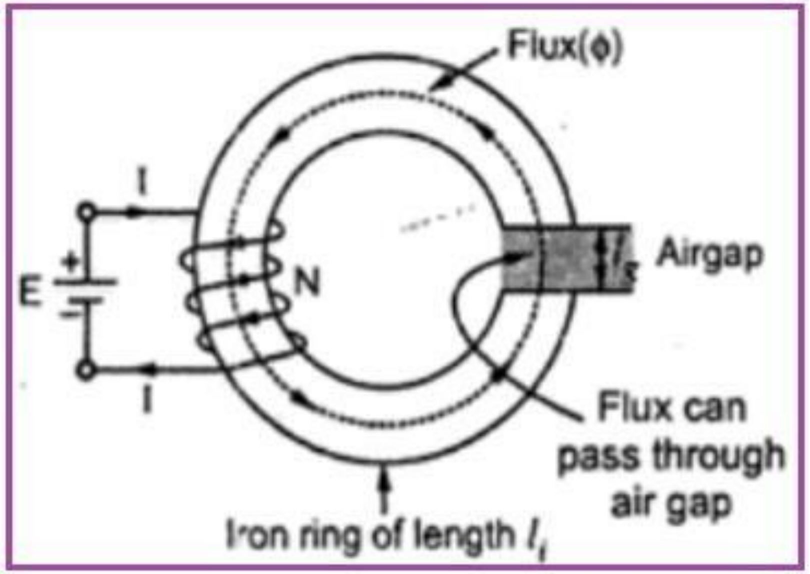
$$\text{Total } S_T = S_1 + S_2 + S_3 = \frac{l_1}{\mu_1 a_1} + \frac{l_2}{\mu_2 a_2} + \frac{l_3}{\mu_3 a_3}$$

where $H_1 = \frac{B_1}{\mu_1}, \quad H_2 = \frac{B_2}{\mu_2}, \quad H_3 = \frac{B_3}{\mu_3}$

$$\text{Total } \phi = \frac{\text{Total m.m.f.}}{\text{Total reluctance}} = \frac{NI}{S_T} = \frac{NI}{(S_1 + S_2 + S_3)}$$

# Series magnetic circuit with air gap

- Consider a ring having mean length of iron part as $l_i$




Total $m.m.f = NI$ AT

Total reluctance $S_T = S_i + S_g$

Where $S_i$=reluctance of iron path

$S_g$=reluctance of air gap

$$S = \frac{l_i}{\mu a_i} \qquad S_g = \frac{l_g}{\mu_o a_g}$$

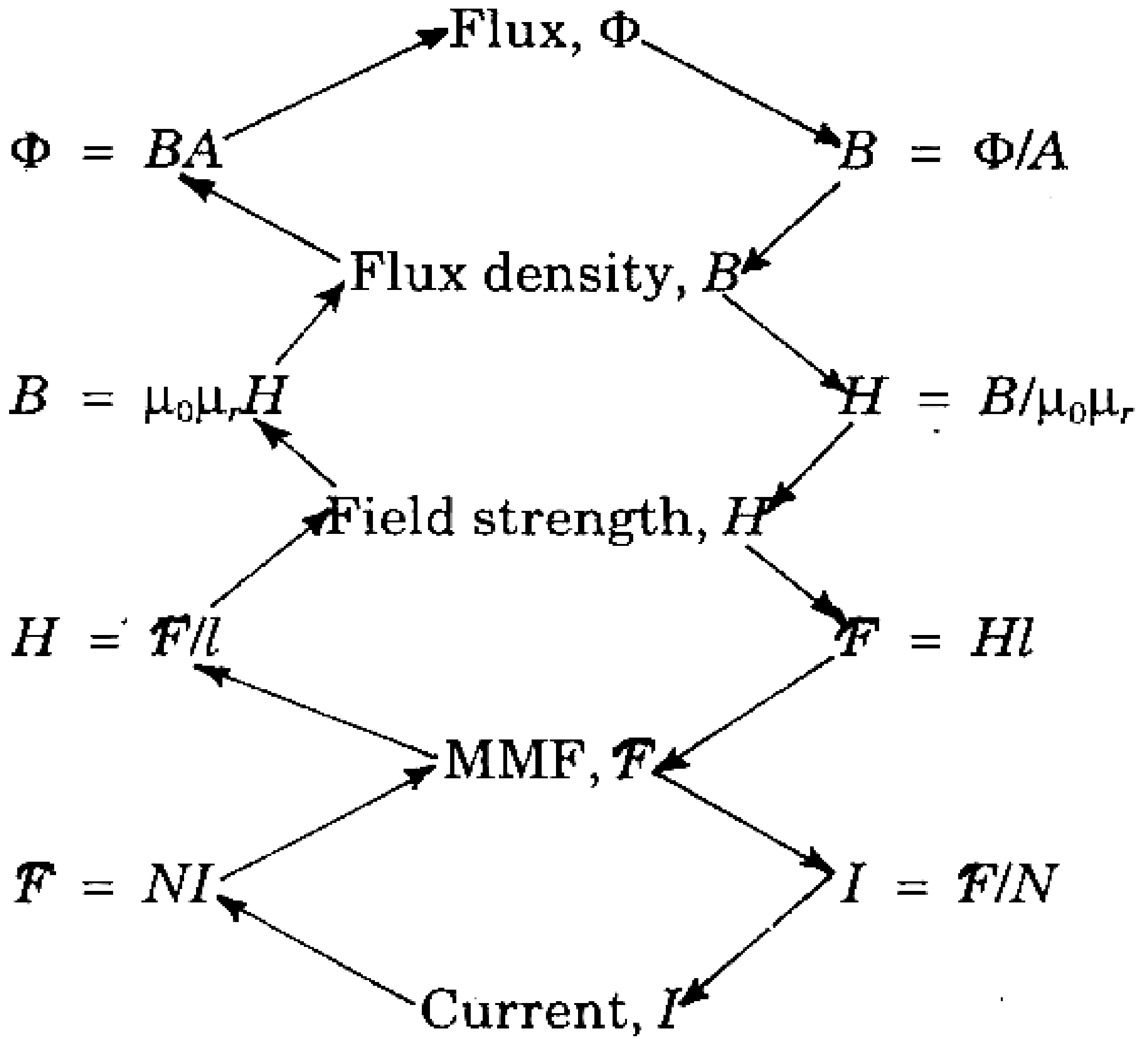
$$S_T = \frac{l_i}{\mu a} + \frac{l_g}{\mu_o a}$$

$$\emptyset = \frac{m.m.f}{reluctance} = \frac{NI}{S_T}$$

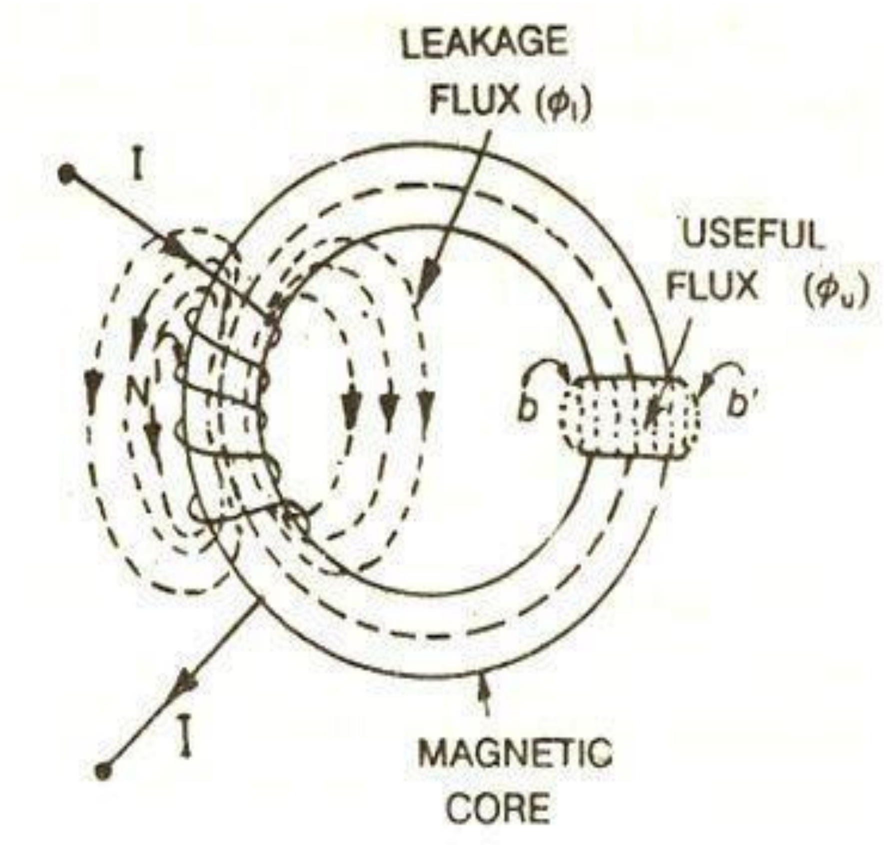
# Kirchhoff's Laws

- ***Kirchhoff's Flux Law* (KFL) :** *The total magnetic flux towards a junction is equal to the total magnetic flux away from that junction.*
- ***Kirchhoff's Magnetomotive Force Law* (KML) :** *In a closed magnetic circuit, the algebraic sum of the product of the magnetic field strength and the length of each part of the circuit is equal to the resultant magnetomotive force.*

# Steps to solve a problem on magnetic circuit



# Leakage Flux and Fringing




- Leakage Flux : the magnetic flux which does not follow the particularly intended path in a magnetic circuit.
- When a current is passed through a solenoid, magnetic flux is produced by it.

- Most of the flux is set up in the core of the solenoid and passes through the particular path that is through the air gap and is utilised in the magnetic circuit. This flux is known as **Useful flux** $\emptyset_u$

- Practically it is not possible that all the flux in the circuit follows a particularly intended path and sets up in the magnetic core and thus some of the flux also sets up around the coil or surrounds the core of the coil, and is not utilised for any work in the magnetic circuit. This type of flux which is not used for any work is called **Leakage Flux** and is denoted by $\emptyset_l$.

- The total flux Φ produced by the solenoid in the magnetic circuit is the sum of the leakage flux and the useful flux.

$$\boxed{\varphi = \varphi_u + \varphi_l}$$

## Leakage coefficient

- The ratio of the total flux produced to the useful flux set up in the air gap of the magnetic circuit is called leakage coefficient or leakage factor. It is denoted by (λ).

$$\lambda = \frac{Total\ flux(flux\ in\ the\ iron\ path)}{useful\ flux(flux\ in\ the\ air gap)}$$

$$\boxed{\lambda = \frac{\varphi}{\varphi_u}}$$

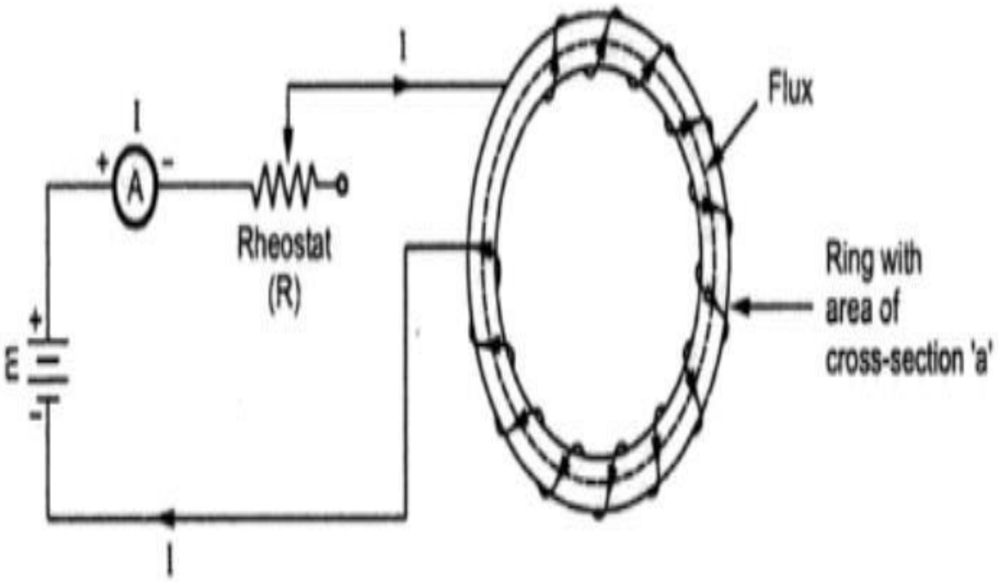
## Fringing

- The useful flux when sets up in the air gap, it tends to bulge outward at (b and b’) as shown in figure, because of this bulging the effective area of the air gap increases and the flux density of the air gap decreases. This effect is known as **Fringing** and the longer the air gap the greater is the fringing.

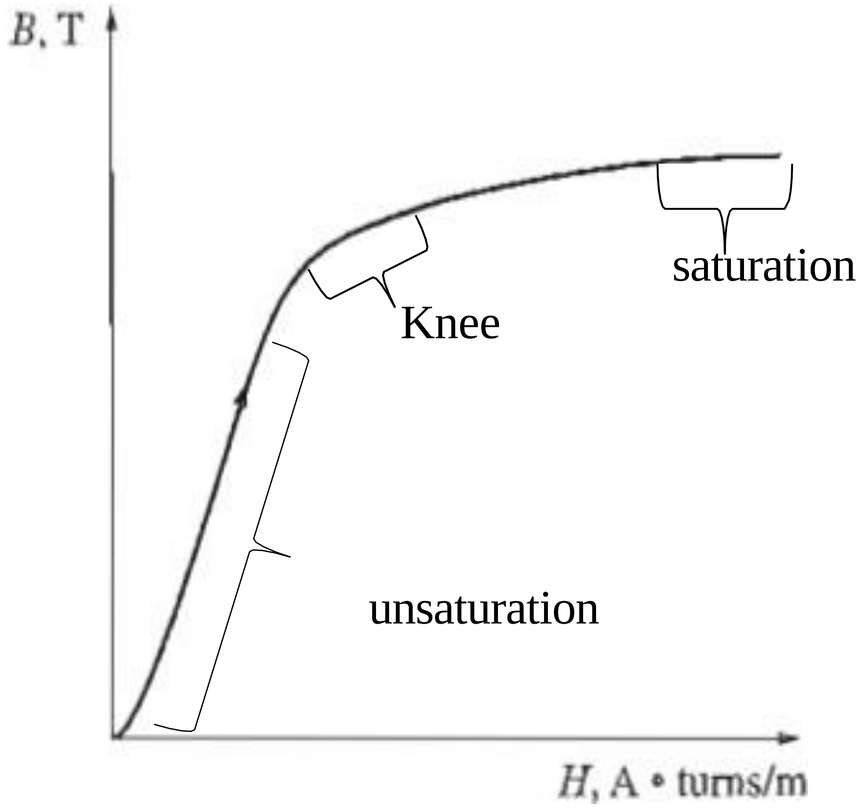
# Magnetic Behavior of Ferromagnetic Materials

- To illustrate the behavior of magnetic permeability in a ferromagnetic material, apply a direct current to the core, starting with 0 A and slowly working up to the maximum permissible current.
- At first, a small increase in the magnetomotive force produces a huge increase in the resulting flux. After a certain point, though, further increases in the magnetomotive force produce relatively smaller increases in the flux. Finally, an increase in the magnetomotive force produces almost no change at all.
- The graph between the flux density(B) and the magnetic field intensity(H) for the magnetic material is called its **magnetization curve** or B-H curve.
- It is also called a **saturation curve.**

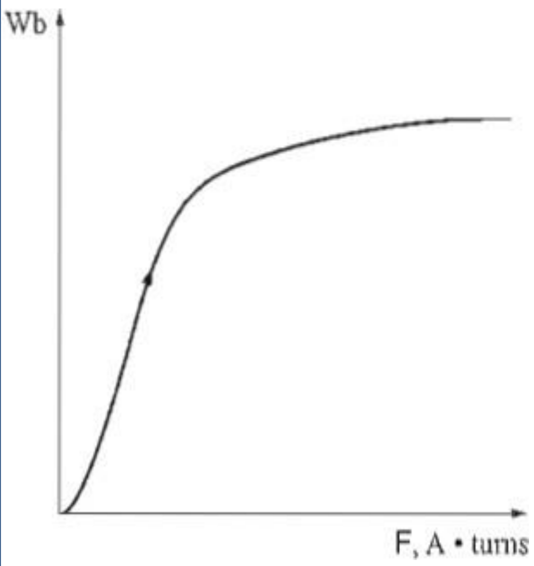
# Cont….

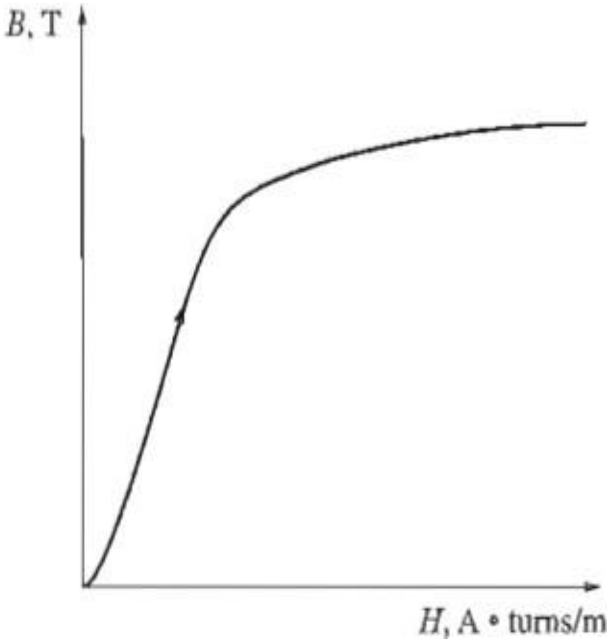


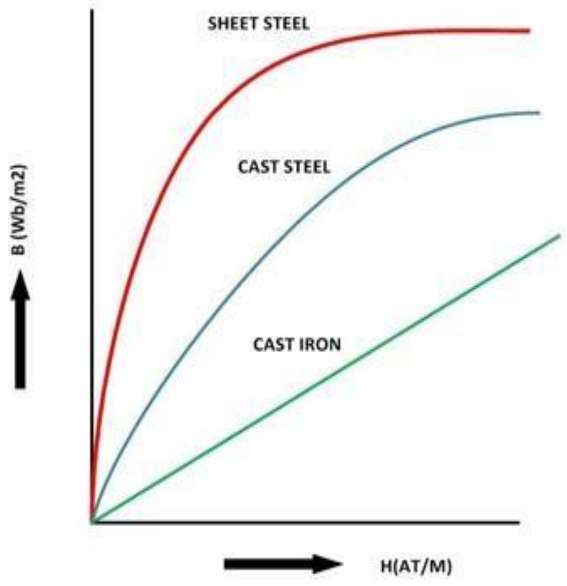

**Experimental set up to obtain B-H curve**



- **Magnetic Saturation** is The limit beyond which magnetic flux density in a magnetic area does not increase sharply further with increase of mmf.
- **Residual magnetism** is the amount of magnetization left behind after removing the external magnetic field from the circuit. In another word the value of the flux density retained by the magnetic material is called Residual Magnetism and the power of retaining this magnetism is called retentivity of the material. or
- **Residual flux density** is the certain value of magnetic flux per unit area that remains in the magnetic material without presence of magnetizing force (i.e. H = 0).




$$H = \frac{Ni}{l_c} = \frac{\mathcal{F}}{l_c}$$

$$\phi = BA$$

The magnetization curve expressed in terms of flux density and magnetic field intensity.

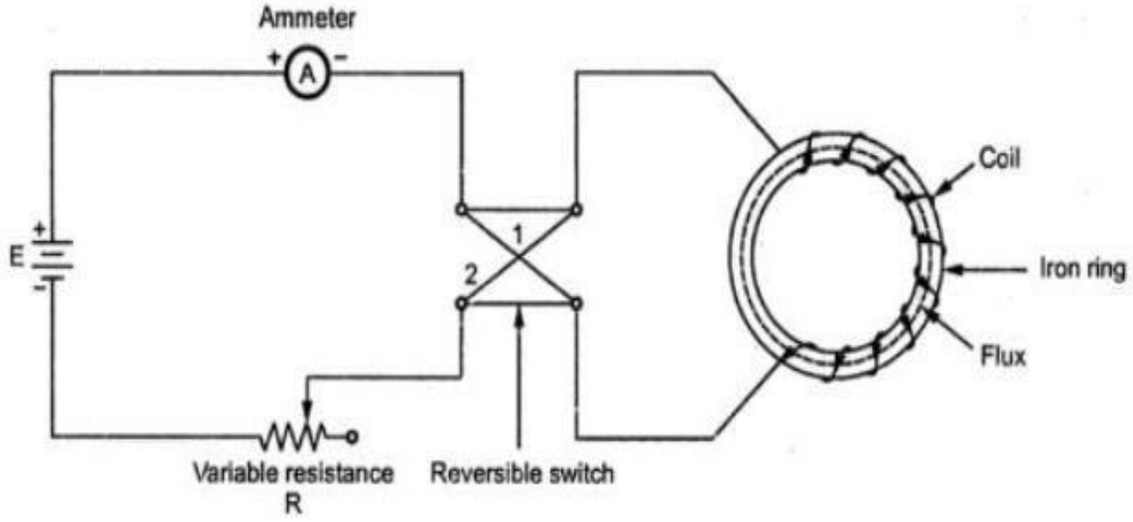
Magnetization curve of different magnetic materials

- The region of this figure in which the curve flattens out is called the *saturation region*, and the core is said to be *saturated.*
- In contrast, the region where the flux changes very rapidly is called the *unsaturated region* of the curve, and the core is said to be *unsaturated.*
- The transition region between the unsaturated region and the saturated region is sometimes called the *knee* of the curve.
- The value of relative permeability mainly depends on the value of flux density. But for the non-magnetic materials like plastic, rubber, etc. and for the magnetic circuit having an air gap, its value is constant, denoted by ($\mu_0$). Its value is $4\pi x10^{-7}$H/m and commonly known as absolute permeability or permeability of free space

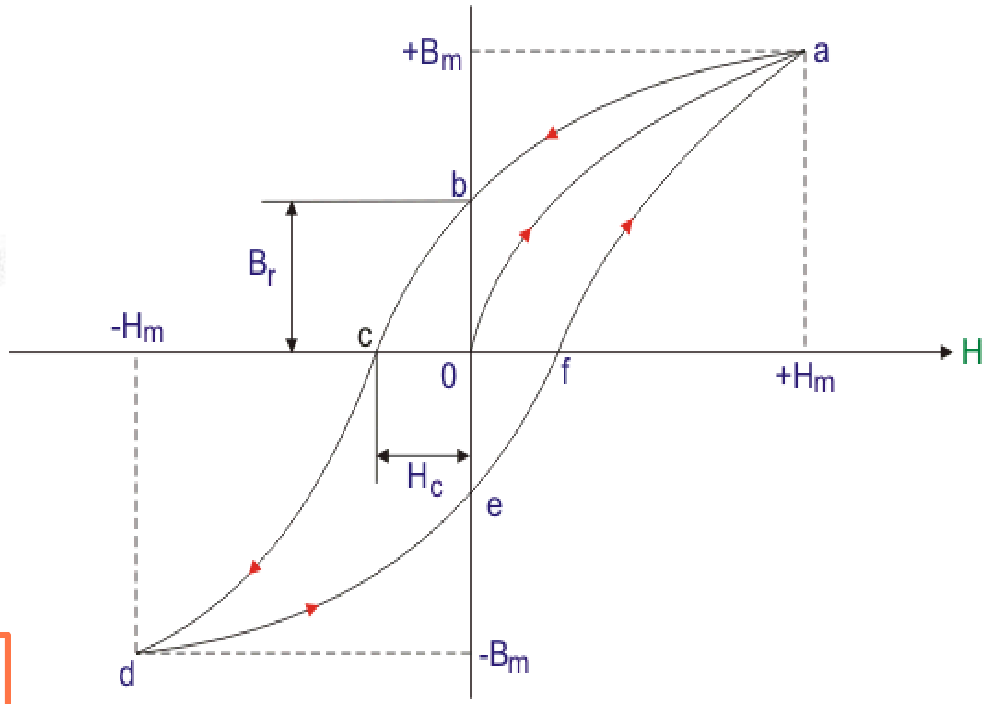
$H_{max}$

# Magnetic Hysteresis

- **1:** When supply current I = 0, so no existence of flux density (B) and magnetizing force (H). The corresponding point is **'O'** in the graph below.
- **2:** When current is increased from zero value to a certain value, magnetizing force (H) and flux density (B) both are set up and increased following the path **o – a.**
- **3:** For a certain value of current, flux density (B) becomes maximum ($B_{max}$). The point indicates the magnetic saturation or maximum flux density of this core material. All element of core material get aligned perfectly. Hence $H_{max}$ is marked on H axis. So no change of value of B with further increment of H occurs beyond point **'a'.**




Experimental set up to obtain Hysteresis Loop




***Hysteresis Loop***

- **4:** When the value of current is decreased from its value of magnetic flux saturation, H is decreased along with decrement of B not following the previous path rather following the curve **a – b.**
- **5:** The point **'b'** indicates H = 0 for I = 0 with a certain value of B. This lagging of B behind H is called **hysteresis.** The point 'b' explains that after removing of magnetizing force (H), magnetism property with little value remains in this magnetic material and it is known as residual magnetism ($B_r$). Here **o – b** is the value of residual flux density due to retentivity of the material.

- **6:** If the direction of the current I is reversed, the direction of H also gets reversed. The increment of H in reverse direction following path **b – c** decreases the value of residual magnetism ($B_r$) that gets zero at point **'c'** with certain negative value of H. This negative value of H is called coercive force ($H_c$).
- **7:** H is increased more in negative direction further; B gets reverses following path **c – d.** At point **'d'**, again magnetic saturation takes place but in opposite direction with respect to previous case. At point **'d'**, B and H get maximum values in reverse direction, i.e. ($-B_m$ and $-H_m$).

- **8:** If we decrease the value of H in this direction, again B decreases following the path **de.** At point **'e'**, H gets zero valued but B is with finite value. The point **'e'** stands for residual magnetism ($-B_r$) of the magnetic core material in opposite direction with respect to previous case.
- **9:** If the direction of H again reversed by reversing the current I, then residual magnetism or residual flux density ($-B_r$) again decreases and gets zero at point **'f '** following the path **e – f.** Again further increment of H, the value of B increases from zero to its maximum value or saturation level at point **a** following path **f – a.**
- The path **a – b – c – d – e – f – a** forms hysteresis loop.

**[NB: The shape and the size of the hysteresis loop depend on the nature of the material chosen]**

- **Hysteresis:** The phenomenon of flux density(B) lagging behind the magnetizing force
(H) in a magnetic material is known as **Magnetic Hysteresis.**
- **Coercive force** is defined as the negative value of magnetizing force (-H) that reduces residual flux density of a material to zero.
- **Retentivity:**It is defined as the degree to which a magnetic material gains its magnetism after magnetizing force (H) is reduced to zero.
- The ***hysteresis loss*** in an iron core is the energy required to accomplish the reorientation of domains during each cycle of the alternating current applied to the core.
- The area enclosed in the hysteresis loop formed by applying an alternating current to the core is directly proportional to the energy lost in a given ac cycle.

# Thought of the DAY

There are no secrets to success.

It is the result of

preparation, hard work,

and learning from

failure.

--Colin Powell..

# Summary

- Current flowing in a conductor creates a magnetic field around it.

- The complete closed path followed by any group of magnetic lines of force is termed as magnetic circuit.

- The characteristics of magnetic circuits are analogous with that of electric circuits.